Deepika Alagiriswamy Panneerselvam

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PROFESSIONAL SUMMARY

Dynamic Data Engineer with 3+ years of experience designing, building, and optimizing high-impact data pipelines across cloud and big-data ecosystems. The specialty includes developing scalable ETL workflows using PySpark and Snowflake, modernizing legacy systems on AWS, automating deployments with CI/CD, and enabling reliable, analytics-ready datasets for enterprise use cases.

EXPERIENCE

Starplot - Chicago, USA

Data Engineer October 2025 - Present

- Supported Starplot's mission of delivering a **unified partnership management** system that helps organizations manage partners and customers in one platform.
- Built a **TAG (Table-Augmented Generation) based system** for the Starplot dashboard, enabling users to query and retrieve insights directly from relational databases.
- **Prepared and structured organizational datasets** for the upcoming Marketplace feature to ensure clean and consistent data flows.
- Contributed to **full-stack development** using **React** and **Laravel** frameworks, enabling faster product updates and a more seamless platform experience.

Urban Institute - Washington DC, USA

Data Engineering Intern June 2024 - November 2024

- **Modernized the 8-year-old** EMR cluster provisioning system by automating version updates for RStudio, R, and EMR via custom APIs, delivering the upgrade two months early while reducing manual effort by 50% and improving workload speed by 25%.
- **Designed and implemented AWS Step Functions** to orchestrate EMR cluster and EC2 instance creation, streamlining automation and reducing setup time from hours to minutes. This enabled the team to accelerate troubleshooting and reduced the manual intervention.
- Built a CI/CD pipeline using **AWS SAM** and **GitHub Actions**, fully automating deployments, cutting the release time by 60%.

Dun & Bradstreet - Hyderabad, India

Data Engineer August 2021 - July 2023

- Improved **6 data pipelines** by adding key business features and restructuring data models, resulting in smoother access for analytics.
- Deployed **3 end-to-end ETL pipelines** into production using **PySpark, Snowflake, AWS** and **Airflow**, improving data delivery.
- Orchestrated workflows using **Apache Airflow** and contributed to **Snowflake data modeling** to streamline pipelines and improve query efficiency, enhancing overall system performance for large-scale data operations.
- **Reduced production failure by 30%** by identifying and fixing high-priority bugs through **debugging** and **performance tuning**.
- Handled diverse file formats—including **nested JSON**, **CSV**, and **Parquet**—within automated **data ingestion** workflows using **AWS Lambda,** enabling seamless integration of high-volume datasets from multiple enterprise sources.
- Set up end-to-end **CI/CD pipelines** using **Jenkins**, ensuring version control and automated testing, enabling faster deployments.
- Created **SQL scripts** to validate data pipelines, ensuring **accuracy**, **compliance**, and **data governance**- reducing manual checks by approximately 40% and strengthening overall data quality across production environments.

Data Engineer Intern February 2021 - August 2021

- **Documented 6 ETL** workflows with **PySpark** and **Snowflake**, improving knowledge transfer and reducing onboarding time.
- Monitored production jobs using **AWS CloudWatch** and **SQL validation**, cutting data issues and potential downtime by ~50%.
- Collaborated closely with **stakeholders** and **senior engineers** to develop scalable pipelines for high-impact, critical tasks.

SKILLS

Programming Languages & Libraries: Python, Advanced SQL, PySpark, Pandas, NumPy, R, Bash scripting, Jupyter Notebooks.
Big Data Technologies & Databases: Apache Spark, Databricks, Hadoop, Snowflake, MySQL.
Cloud Services: AWS - Lambda, EMR, Step Functions, SAM, S3, CloudFormation.
GCP – BigQuery, Cloud Functions, GCS, Dataproc, Pub/Sub.
Automation & Orchestration: Apache Airflow, Jenkins, Git, GitHub, GitHub Actions, Jira, Docker
Data Visualization: Microsoft Excel, Tableau, Superset.

PROJECTS

Stock Market Data Pipeline using Kafka and AWS March 2025

- Designed a real-time pipeline in Python to simulate live trading data, stream via Kafka, and store securely in S3 buckets.
- Used AWS Glue to crawl and catalog S3 data, enabling fast, structured querying and analysis using Amazon Athena with ease.
- **Tech Stack:** Python, Pandas, Apache Kafka (on AWS EC2), AWS S3, AWS Glue (Crawler + Data Catalog), AWS Athena.

Multilabel Predictions on Academic Articles, Illinois Tech, Chicago, IL April 2024

- Multi-label classification model that processes academic articles using TF-IDF vectorization and BERT to train the models for predicting article's subjects.
- Model performance is evaluated with precision, recall, and F1-score metrics; Matplotlib visualizations to understand label distribution and EDA.
- **Tech Stack:** Machine Learning, NLP, Python, Pandas, NumPy, BERT, scikit-learn, Matplotlib.

Predictions for the 2023 ICC Cricket World Cup, Illinois Tech, Chicago, IL November 2023

- Built an R-based machine learning model to predict ICC World Cup 2023 outcomes using historical team data and visualized insights.
- Trained Random Forest models including team stats, venue, toss, pitch data to improve prediction accuracy and performance of match outcomes.
- **Tech Stack:** R, Random Forest, ggplot, Tidyverse.

Heart Attack Risk Prediction, Illinois Tech, Chicago, IL April 2024

- Built a predictive model to assess the risk of heart attack based on patient health metrics, enabling early identification of high-risk individuals.
- Leveraged statistical analysis and supervised learning techniques to uncover key risk factors, improving the interpretability and accuracy of predictions.
- This project demonstrates the potential of data-driven approaches in advancing preventive healthcare and personalized risk assessment.
- **Tech Stack:** Python, Pandas, Scikit-learn, Data Analytics, Machine Learning.

EDUCATION

Illinois Institute of Technology, Chicago, IL May 2025

Master of Data Science, GPA: 3.90

- Coursework: Data Preparation and Analysis, Big Data Technologies, Machine Learning, Regression, Statistics, Advance Database Organization.

Coimbatore Institute of Technology, Coimbatore, India May 2021

Bachelor of Technology, Information Technology, GPA: 3.80

- Coursework: Data Structures, Database Management System, Object Oriented Programming, Cloud Computing, Data Mining and Warehousing.

CERTIFICATIONS

- AWS Certified Data Engineer – Associate *(August 2024)*
- Google Cloud Certified Cloud Digital Leader *(February 2025)*
- AWS Certified AI Practitioner *(September 4, 2025)*